THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(d)** 3 x 10^5	2 x 10^30 / 6 x 10^24.
15	**(b)** 50 per cent	30/60.
16	**(c)** 2 : 1	75/25 = 3; 60/40 = 1.5; ratio 2 : 1.
17	**(d)** S^{2-}	Z = 16 is sulphur, 2 extra electrons.
18	**(c)** u / sqrt(2)	v squared = u squared - 2g(h/2) with h = u squared/2g.
19	**(c)** 1 : 4	a is inversely proportional to m.
20	**(c)** 100 J	Half converted to PE.
21	**(d)** 0.68 m and 3 m	lambda = v/f.
22	**(c)** hypotonic to the cell	Water entered.
23	**(a)** accumulation of food	Phloem blocked.
24	**(d)** Fungi	Hyphae and chitin.
25	**(d)** 14 March	Mid cycle.
26	**(c)** 4000 J	400 W x 10 s.
27	**(d)** the theory is strongly supported but remains open to tests	Science is provisional.
28	**(a)** only one component sublimes	Different behaviour on heating.
29	**(a)** 4.4 g	Mass of gas escaped.
30	**(b)** 24.3 u	(1896 + 250 + 286)/100 = 24.32.
31	**(c)** 8 km/h	24 km/ 3 h.

32	**(a)** the pull of the book on the Earth	Third-law pair.
33	**(b)** 250 J	mgh.
34	**(d)** 24 m/s	0.6 x 40.
35	**(a)** synthesis of ATP	Mitochondria do ATP.
36	**(b)** glandular epithelium	Secretion.
37	**(b)** They are in the same class and phylum	Higher ranks shared.
38	**(a)** the one with an eye grows	Bud needed.
39	**(c)** algal bloom and fish kill	Eutrophication.
40	**(d)** A is false but R is true.	A is false (theories can be refined); R is true.
41	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
42	**(c)** A is true but R is false.	A is true; R is false because the gas escapes.
43	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
44	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
45	**(a)** 4 m/s squared	F/m.
46	**(b)** 480 W	2400/5.
47	**(d)** 150 m	1500 x 0.2/2.
48	**(c)** cell wall	Rigid wall.
49	**(a)** sensory neuron, relay neuron and motor neuron	Reflex arc.
50	**(b)** animal	Animalia.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** wings or hairs	Light structures.
2	**(b)** the ground is heated by sunlight and warms the air	Surface heating.
3	**(d)** 5.4 cm	Subtract the zero error.
4	**(a)** 32 g	Water 100 g holds 32 g; crystals 64 - 32.
5	**(b)** 3 : 8, CO_2	Atoms ratio 2.4/12 : 6.4/16 = 1 : 2.
6	**(d)** 20 and 16	X lost 2, Y gained 2.
7	**(b)** 1.5 t	v = at; retard time = at/2a = t/2.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** lignin	Lignified walls.
13	**(a)** dolphins have lungs, hair and mammary glands	Mammalia vs Pisces.
14	**(b)** oviduct	Tube.
15	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
16	**(d)** dosage of a medicine for a child	Safety critical.
17	**(b)** iron and sulphur	Only iron is magnetic.
18	**(d)** 78.5	52.5/47.5 x 71 = 78.5.
19	**(d)** 20	10x + 11(100 - x) = 1080.
20	**(a)** 4.8 km/h	Harmonic mean.
21	**(c)** 20 m/s	a = 4 m/s squared; v = 20 m/s.
22	**(c)** 3h/4	KE = 1/4 of mgh.
23	**(c)** 2 s	680/340.
24	**(a)** hypertonic to A and hypotonic to B	Different internal concentrations.
25	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
26	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
27	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
28	**(b)** increase	More absorption.
29	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
30	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
31	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.

32	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
33	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
34	**(b)** reducing the friction by rollers	Reduce friction.
35	**(a)** 4 m/s	sqrt(2 x 10 x 0.8).
36	**(c)** 86 m	vt.
37	**(b)** bacterial	Prokaryote.
38	**(b)** xylem is intact	Water still moves.
39	**(b)** Gymnosperms	Cones with naked seeds.
40	**(d)** 13 to 15 March	Around ovulation.
41	**(d)** water filtration and flood control	Natural sponge.
42	**(d)** 1 per cent	0.2/20.4.
43	**(d)** 18 g	50 - 32.
44	**(b)** 28 g	28/44 of the mass is nitrogen.
45	**(d)** argon	18 electrons.
46	**(b)** 2t	t proportional to sqrt(h).
47	**(a)** 200 N	0.1 x 20 / 0.01.
48	**(b)** 3 : 20	0.3 : 2.
49	**(c)** 340 m/s	34/0.1 = 340 - note: for illustration.
50	**(b)** large central vacuole	Takes up space.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** a single nucleus and spindle shape	Unstriated.
2	**(c)** belong to the same genus but different species	Same genus Canis.
3	**(a)** cambia of stock and scion are in contact	Tissue contact.
4	**(a)** absorption of UV by ozone	Ozone heating.
5	**(a)** check the experiment and the assumptions in the model	Investigate.
6	**(c)** dissolve in water, filter, evaporate	Sugar dissolves.
7	**(b)** 27.3 per cent	12/44 x 100 = 27.3.
8	**(a)** electron	An electron is about 1/1836 of the mass of a proton.
9	**(a)** 24 m/s	2 x 20 x 30/50.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(c)** A is true but R is false.	A is true; R is false: KE is zero there.
12	**(d)** A is false but R is true.	A is false (loudness does not depend on frequency); R is true.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(c)** dendrite to axon	Direction.
15	**(b)** Plantae	Plant cell.
16	**(b)** hooks or fleshy fruit	Burrs.
17	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
18	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
19	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
20	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
21	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
22	**(d)** A is false but R is true.	A is false: acceleration is g throughout; R is true.
23	**(d)** 10 m/s squared downward	Only gravity acts.
24	**(b)** 4 : 1	KE = p squared/2m.
25	**(c)** 1 : 3	Distance proportional to time.
26	**(a)** 10	Mitosis keeps 20; meiosis halves.
27	**(c)** safranin	Red lignin.
28	**(d)** have hair and mammary glands	Mammals.
29	**(d)** prostate and seminal vesicles	Semen.
30	**(c)** 100	2200/22.
31	**(a)** 15 per cent	3 x 5 per cent.
32	**(a)** the one with R_f 0.9	Largest R_f.
33	**(c)** 310 u	120 + 62 + 128.
34	**(d)** 5	Z = 15: 2, 8, 5.
35	**(d)** 1 : 3 : 5	Odd numbers.

36	**(a)** the same for both	F = change in p/t.
37	**(a)** 72 J	F = m(g + a) = 24 N.
38	**(b)** 20 m	t1 = sqrt(2h/g), t2 = h/v: h = 20 gives 2 + 0.06.
39	**(a)** energy release	Respiration.
40	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
41	**(c)** A is true but R is false.	A is true; R is false: they are heterotrophic.
42	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
43	**(b)** its increase beyond natural levels enhances the greenhouse effect	Balance matters.
44	**(a)** to test the procedure and find problems	Trial run.
45	**(a)** distillation	Boiling point.
46	**(d)** 4 g	36 x 2/18 = 4 g.
47	**(d)** 207	Sum.
48	**(c)** 31.25 m	t up = 2.5; h = 0.5 x 10 x 6.25.
49	**(a)** 10 N	20/2.
50	**(d)** 2520 rupees	360 x 7.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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